Voltage regulation and voltage instability

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Maximum power transfer in a DC system

2. The nose curve in an AC system

3. Examples of voltage instabilities

4. Some counter measures

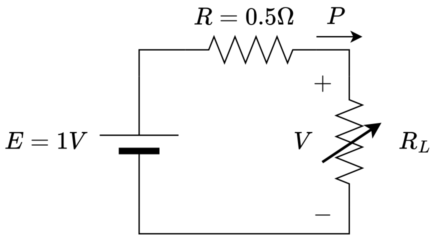
5. Impact of renewable energy resources (RES)

6. Recap

Maximum power transfer in a DC system

Maximum power transfer: reminder I

Consider the circuit below, where the DC source E feeds a load R_L that can be adjusted, e.g. to consume power for heating purposes. R models the internal resistance of the source and the losses in the connecting lines.



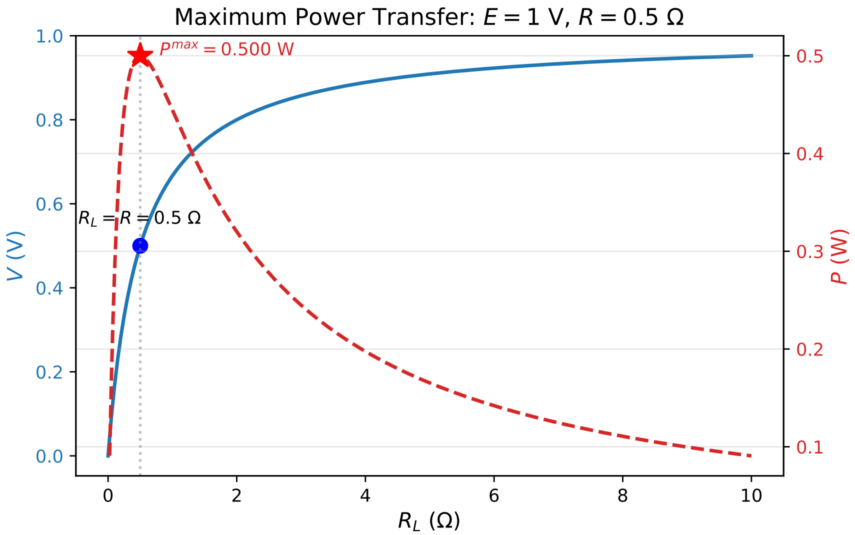
The power transferred to the load is given by:

$$P = R_L I^2 = R_L \left(\frac{E}{R + R_L} \right)^2$$

To find the maximum power transferable to the load, we set the partial derivative of P with respect to R_L to zero and find that $R_L = R$. The maximum power transferable to the load is then:

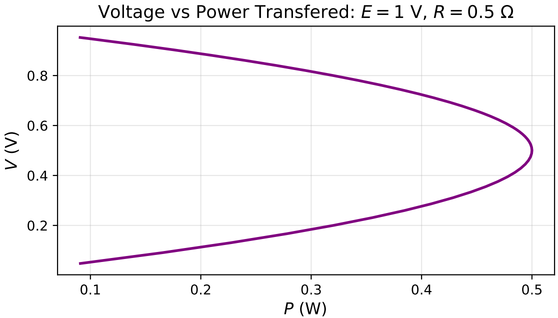
$$P^{max} = \frac{E^2}{4R}$$

Maximum power transfer: reminder II



Voltage vs Power transferred: the nose curve

If we now plot the voltage V across the load as a function of the power P transferred to the load (i.e. for several values of R_L), we obtain the following "**nose curve**":



Exercise

Show that the relationship between V and P is given by $P = \frac{VE - V^2}{R}$

Observations on the nose curve

- there are two possible voltages for a given power transfer, except at the maximum power point where there is only one solution.
- the top part corresponds to $R_L > R$ (the load is larger than the source resistance) while the bottom part corresponds to $R_L < R$.
- the top part has higher voltage and lower current, hence lower losses in the source resistance: **this is where we want to operate!**

Illustration of a voltage instability mechanism

We consider the same circuit as before, but **now the load resistance R_L is controlled** according to the following differential equation:

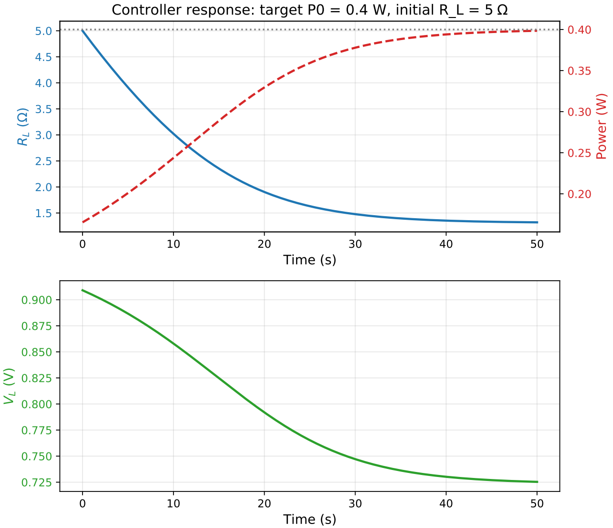
$$\frac{dR_L}{dt} = R_L I^2 - P_0$$

where P_0 **is a target power to be consumed by the load** (**the demand**). The idea is that if the power consumed by the load is below the target power P_0, then R_L decreases, which tends to increase the power consumed by the load, and *vice versa*.

Let us simulate this system for different values of P_0 (the dynamics will also change as a function of the initial conditions $R_L(0)$, obviously, but this will not change the conclusion).

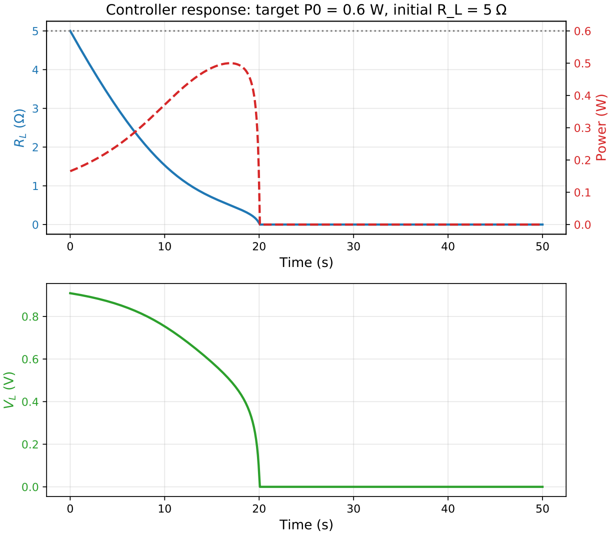
Stable case

If $P_O = 0.4W$, i.e. below the maximum power transferable, the voltage converges to a stable operating point (although it is below 90% of the source voltage):



Unstable case: voltage collapse

If $P_O = 0.6W$, i.e. exceeding the maximum power transferable, we have:



Take home messages

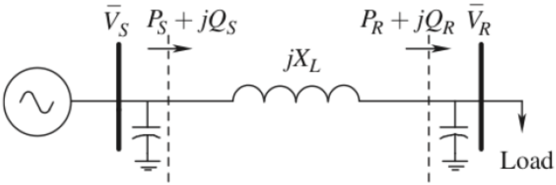
- This (simple) regulator of the load does not know about the maximum power transferable through the source and the connecting lines.
    - This depends on the network characteristics.
    - It thus does not know whether it operates in a stable or unstable region.
- Even if the case is stable from a voltage perspective,
    - the voltage may be low and thus the current high, potentially leading to thermal overloads,
    - an incident on the network (e.g. a line tripping) may reduce the maximum power transferable below the target power P_0, leading to voltage instability and collapse.

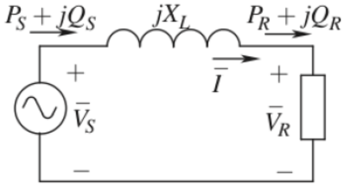
We will see now how this concept applies to AC power systems, and what is the impact of reactive power.

The nose curve in an AC system

Consider the following simple 2-bus AC system I

In an AC system, the resistance is negligible compared to the reactance: $R_L \ll X_L$ (Pay attention that here the index L refers to the line parameters).





Consider the following simple 2-bus AC system II

Remember we established, for $\delta = \delta_S - \delta_R$, that we have:

$$P_R = P_S = \frac{V_S V_R}{X_L} \sin \delta$$

$$Q_R = \frac{V_S V_R \cos \delta}{X_L} - \frac{V_R^2}{X_L}$$

$$Q_S = \frac{V_S^2}{X_L} - \frac{V_S V_R \cos \delta}{X_L}$$

Concept of nose curve I

We **will first consider a load with unity power factor**, i.e. $Q_R = 0$. Hence,

$$\frac{V_S V_R \cos\delta}{X_L} - \frac{V_R^2}{X_L} = 0 \Rightarrow V_S \cos\delta = V_R.$$

Substituting this result in the expression of P_R gives:

$$P_R = \frac{V_S^2}{X_L} \sin\delta \cos\delta = \frac{V_S^2}{2X_L} \sin(2\delta)$$

We can determine the maximum transmissible power by setting the partial derivative to 0:

$$\frac{\partial P_R}{\partial \delta}(\delta^{max}) = \frac{V_S^2}{X_L} \cos(2\delta^{max}) = 0 \Rightarrow \delta^{max} = \frac{\pi}{4}$$

Concept of nose curve II

Replacing δ by δ^{max} in the equation of P_R, we have:

$$P_R^{max} = \frac{V_S^2}{2X_L}$$

and

$$V_R \approx 0.7V_S$$

question

How can we increase the maximum transmissible power through a line?

Concept of nose curve III

One can derive a relationship such that $\frac{V_R}{V_S} = f(\frac{P_R X_L}{V_S^2})$. Consider $y = \frac{V_R}{V_S}$ and $x = \frac{P_R X_L}{V_S^2}$, one has (trust me):

$$y = \sqrt{\frac{1}{2} \pm \sqrt{\frac{1}{4} - x^2}}$$

We verify that y has a unique solution when $x = \frac{1}{2} \Rightarrow P_R = \frac{V_S^2}{2X_L}$, which is the nose of the curve. There is no solution for $P_R > \frac{V_S^2}{2X_L}$ and two solutions for $P_R < \frac{V_S^2}{2X_L}$.

Load characteristics and operating point I

In real life, loads are not necessarily constant impedances. We can consider a variety of load characteristics:

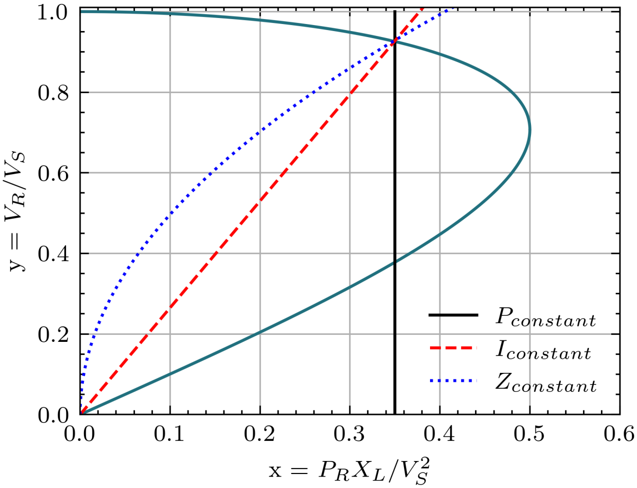
$$P_{constant} \Rightarrow P_R = C_P \Rightarrow x = c_P$$

$$I_{constant} \Rightarrow \frac{P_R}{V_R} = C_I \Rightarrow y = c_I x$$

$$Z_{constant} \Rightarrow \frac{P_R}{V_R^2} = C_Z \Rightarrow y = c_Z \sqrt{x}$$

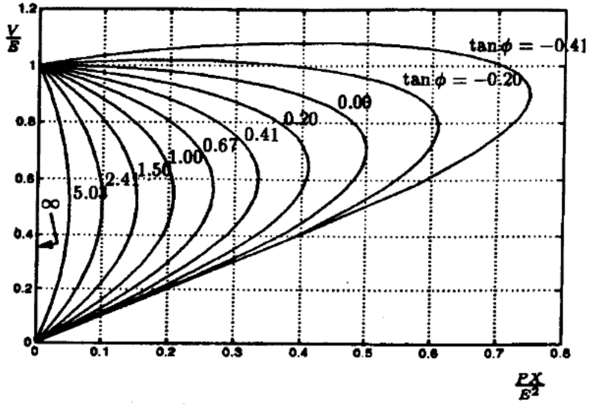
The operating point is where the load characteristic crosses the *nose curve*.

Load characteristics and operating point II



What if the power factor is not unity? I

When you consider $\cos\phi \neq 1$ (the load consumes or produces reactive power), you get more complex nose curves [1].



What if the power factor is not unity? II

The ideas previously developed still stand, except that now the load can:

- *improve* the voltage profile by providing reactive power and thus compensating the reactive power consumed by the line,
- or *worsen* it by further consuming reactive power.

Key points

- There's a maximum transmissible power ($P_R^{max} = \frac{V_S^2}{2X_L}$ for load with $\cos\phi = 1$).
- Inductive character of the line which consumes reactive power (influence of X_L)

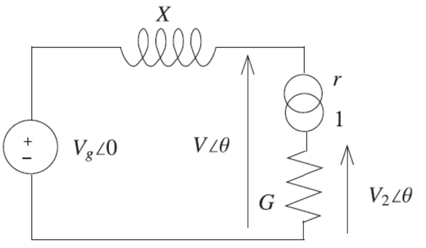
How to increase P_R^{max}?

- By increasing the source voltage V_S,
- By decreasing X_L (adding lines in parallel),
- By producing reactive power at the load side to compensate for the reactive power consumed by the line.

Examples of voltage instabilities

Long-term instabilities I

Consider the circuit on the right. On-load tap changers (OLTCs) change the turn ratio of the transformers feeding the distribution systems to regulate the voltages on the secondary side (V_2).



- The primary side of the transformer is the high voltage network, and the secondary side is the medium voltage network. We assume an ideal transformer: $\frac{V}{V_2} = r$, $\frac{I_2}{I} = r$.
- The load on the secondary side is represented by a constant conductance G, consuming active power.

Long-term instabilities II

- The voltage on the primary side V_g is kept constant by a synchronous generator, as long as its reactive power limits are not reached.

The load characteristic seen from the primary side is $P_G = G\left(\frac{V}{r}\right)^2$, with P_G the power consumed by the conductance G.

Now, imagine one wants to keep $V_2 = V_2^o$, if $V \searrow \Rightarrow r \searrow$. Indirectly, by decreasing r, the OLTC tries to restore the load (since it increases V_2 and $P_G = GV_2^2 = G\left(\frac{V}{r}\right)^2$). Two different scenarios:

- 1) $\frac{V}{r}$ converges towards V_2^o, the load is restored.
- 2) $\frac{V}{r}$ never converges towards V_2^o and V collapses.

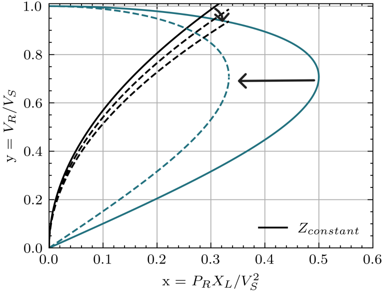
Long-term instabilities III

Imagine a disturbance leading to a decrease in the maximum transmissible power. It can be caused by:

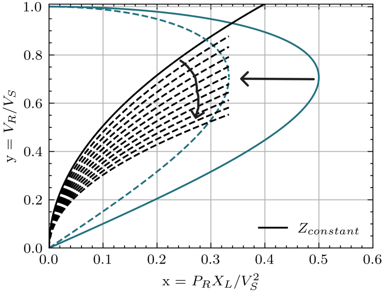
- the tripping of a line (as it is the case here), which leads to $X_L \nearrow$
- or hitting the reactive power limits of a generator, thus V_S is no longer maintained.

Long-term instabilities IV

Two scenarios can happen:



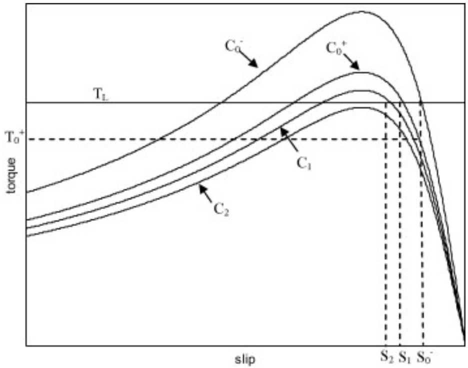
1) P_G is recovered after two tap changes.



2) P_G is never recovered, and the voltage collapses. This is a typical case of **Long-term Voltage Instability**.

Short-term instabilities

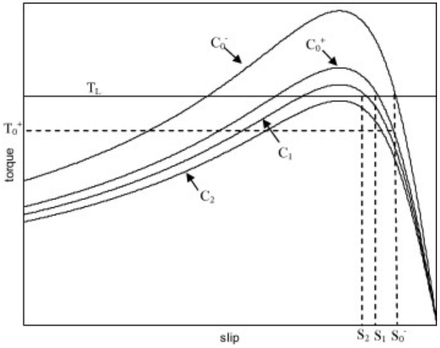
Consider an induction motor. The torque-slip curve is given below.



- T_L is the mechanical torque.
- C curves correspond to different electromechanical torques depending on the slip s, which is related to synchronous speed ω_s and actual rotation speed ω_r by $s = \frac{\omega_s - \omega_r}{\omega_s}$.
- The initial curve is C_0^- and the machine operates on operating point S_0.

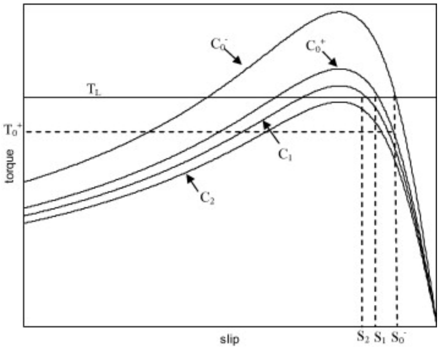
If the voltage drops

- the new curve becomes C_0^+
- the motor speed ω_r is reduced (thus s increases), and we reach a new operating point S_1
- this increase in s leads to an increase in motor current and the further decrease in voltage and electrical torque
- if the rate of decreasing voltage is slower than that of increasing slip, the voltage settles down.

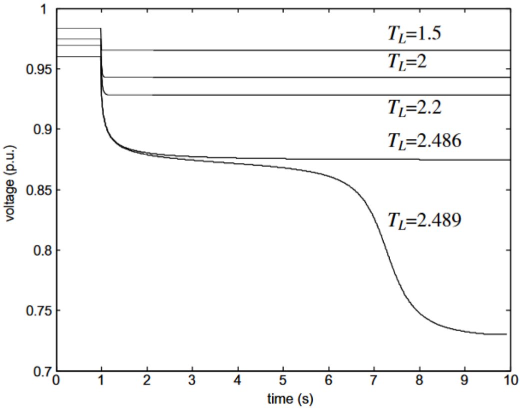


If now the voltage drops faster I

- e.g. the new curve becomes C_2,
- the motor speed is reduced until it completely stops (since there is no intersection between C_2 and T_L).
- The induction motor acts as a large inductance, drawing reactive power.
- This is considered a **Short-term Voltage Instability** as this phenomenon is much quicker than what we have with OLTCs (it takes several seconds to change tap positions).
- OLTCs are not able to restore the voltage.



If now the voltage drops faster II



Some counter measures

Network Reinforcement

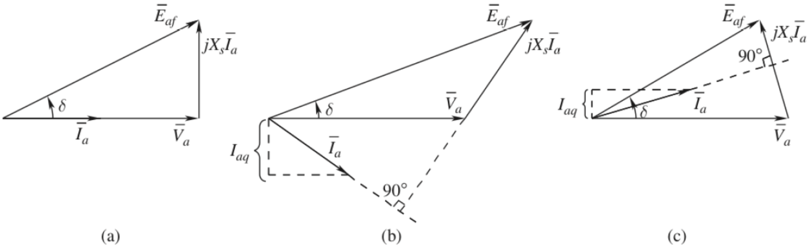
We saw that the main issues for voltage problems are:

- Large values of X_L
- Not enough reactive power compensation

To mitigate voltage problems, one solution is to reinforce the network by adding new lines and capacitor banks, static var compensators or synchronous condensers.

Voltage regulation (reminder) I

The second solution is to use the reactive power reserve already available. In transmission systems, we have synchronous machines. They can provide or consume reactive power.



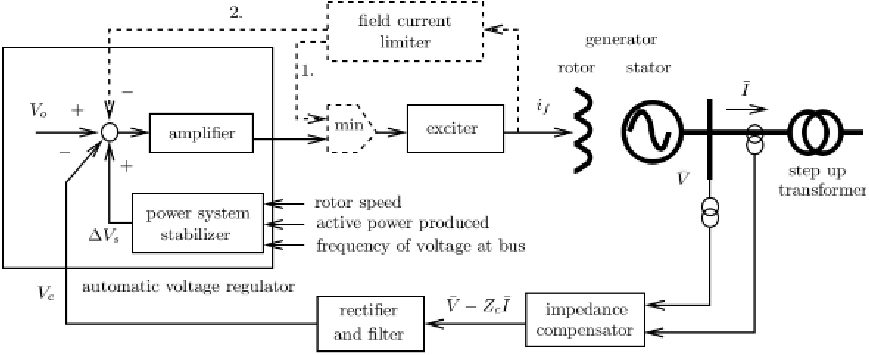
Voltage regulation (reminder) II

In this figure E_{af} is the induced emf, V_a the stator voltage, I_a the stator current and X_s the synchronous reactance. In case (a), there is no reactive power transfer. In case (b), the synchronous machine **provides reactive power** and in case (c), the machine **consumes reactive power**.

- When the machine is overexcited, it produces reactive power (because of the larger induced emf).
- When it is underexcited, it consumes reactive power.

A regulator is responsible for controlling the excitation current in a synchronous generator.

Voltage regulation (reminder) III



Automatic Voltage Regulator (AVR)

The voltage setpoints V_0 are dispatched by the TSO to ensure a safe and reliable grid.

Primary, secondary and tertiary voltage control I

- When a disturbance occurs, or following a change in load (cf. ’duck curve’), the *primary* voltage control loops maintain suitable voltage levels close to the large power plants equipped with AVRs.
  - However, voltages at other buses may move out of tolerance intervals (in either direction), and reactive power reserves may not be shared in an even way among generators.
- *Secondary* voltage control loops can be used at the zonal level, to adjust the set-points of AVRs so as to control the voltage at ’pilot nodes’ in the network while distributing the required reactive power evenly among generators.

Primary, secondary and tertiary voltage control II

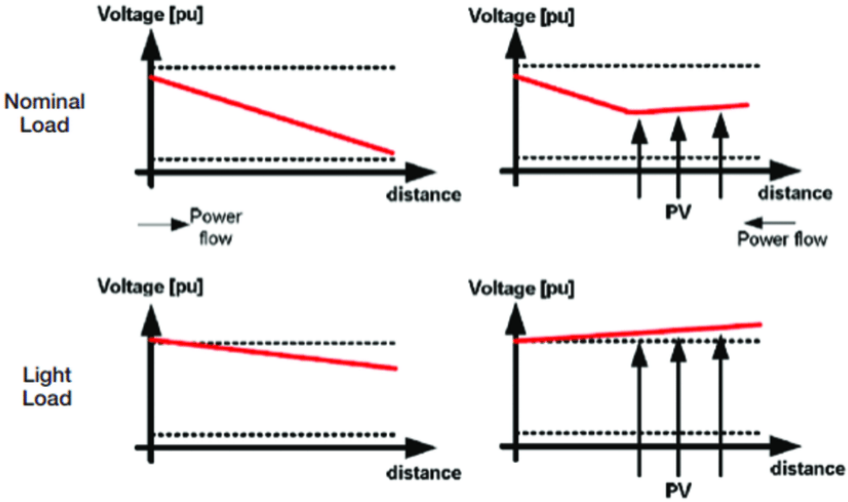
  - Secondary voltage control loops can also be used to switch shunt reactive compensation devices (capacitors/inductors) in order to increase reactive power generation margins in their zone (among a few large power plants).
- *Tertiary* voltage control uses OPF solvers to calculate set-points at pilot nodes and possibly adjust some transformer ratios, so as to minimize losses and maximize MVar reserves at the entire system level.
- Response times of different levels of voltage control
  - *Primary:* 1-3 seconds ; *Secondary:* 30 seconds -3 minutes ; *Tertiary:* 10-15 minutes

Impact of renewable energy resources (RES)

Reverse power flows in distribution systems I

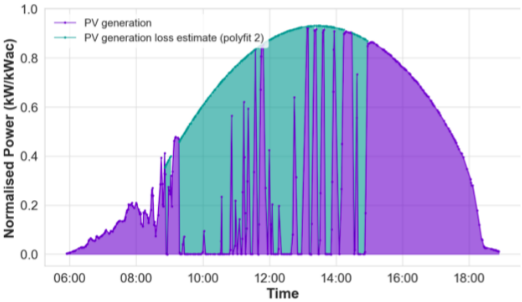
- Distribution networks are radial networks (they are built as meshed networks, but operated radially)
- Before the advent of distributed energy resources (*PV panels, batteries,...*), the power was flowing from the substation node, down to the end of the feeders.
- Increasing penetration of DERs led to reverse power flows.

Reverse power flows in distribution systems II



Reverse power flows in distribution systems

- During hours of high production and low consumption (at midday on a summer day), overvoltages can occur.
- It leads to the disconnection of PV inverters $\Rightarrow$ loss of production.



Reverse power flows in distribution systems

- In distribution systems, $X/R < 1$. Active power flows have actually a greater influence on voltage magnitudes than reactive power flows.
- One way to mitigate overvoltages in distribution networks with high penetration of solar inverters is to do active power curtailment and reactive power compensation.

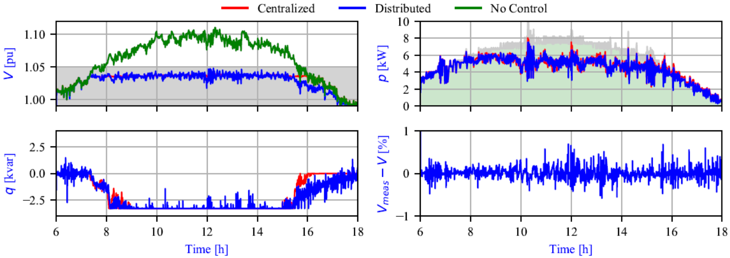
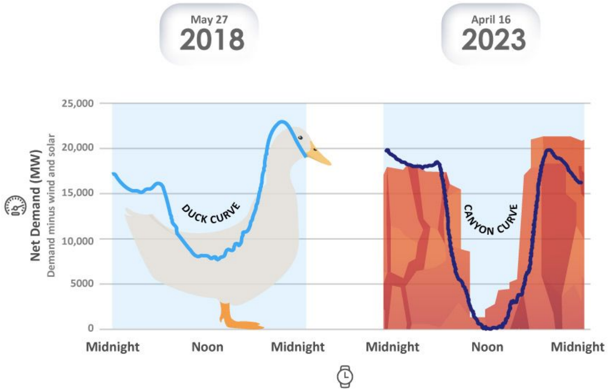


Fig. 3. Results for one specific bus equipped with a PV inverter.

Duck Curve I

- As we further increase the penetration of RES, changes appear in the demand profile (concept of Duck Curve, now becoming Canyon Curve)
- Needs for flexibility to ensure power balance: shut down of flexible production plants when RES start to produce, activate them again when they stop producing
- There exist various solutions to *fill the curve*: Demand Side Management, Energy Buffers (e.g. batteries), increasing import/export capacities.

Duck Curve II



Recap

What did we learn?

- There is a maximum transmissible power through a line.
- Different voltage instabilities, caused by slow-acting devices like OLTCs (long-term instabilities), or by faster dynamics like stalling of induction motors (short-term instabilities).
- To prevent voltage instabilities, one can reinforce the grid by adding new lines or new devices, or enforce voltage regulation (local voltage control tracking voltage setpoints given by the system operator).
- The increasing penetration of RES creates overvoltage problems in distribution networks and challenges for balancing the system.

Principle of Automatic Voltage Control

- *The main tool:* primary voltage control via *Automatic Voltage Regulators (AVRs)* of large synchronous generators and synchronous condensers
- Secondary voltage control and automatic switching of reactive compensation devices and transformer taps
- Tertiary voltage control and voltage profile optimization

Control resources

Which of these control resources are the main levers for voltage stability?

- ☐ Adjust synchronous generators' field current
- ☐ Adjust synchronous generators' mechanical power
- ☐ Change transformer taps
- ☐ Change shunt compensation
- ☐ Act on topology: switch lines and transformers in/out of service
- ☐ Fast start-up generator units
- ☐ In extremis load curtailment
- ☐ Control renewable generation (e.g. PV curtailment)
- ☐ Use batteries and other energy storage systems

References I

T. Van Cutsem and C. Vournas, *Voltage stability of electric power systems*. Springer Science & Business Media, 2007.